



Beyond Entangled Edits: How to truly **separate** and **control** local & global features in pre-trained GANs?

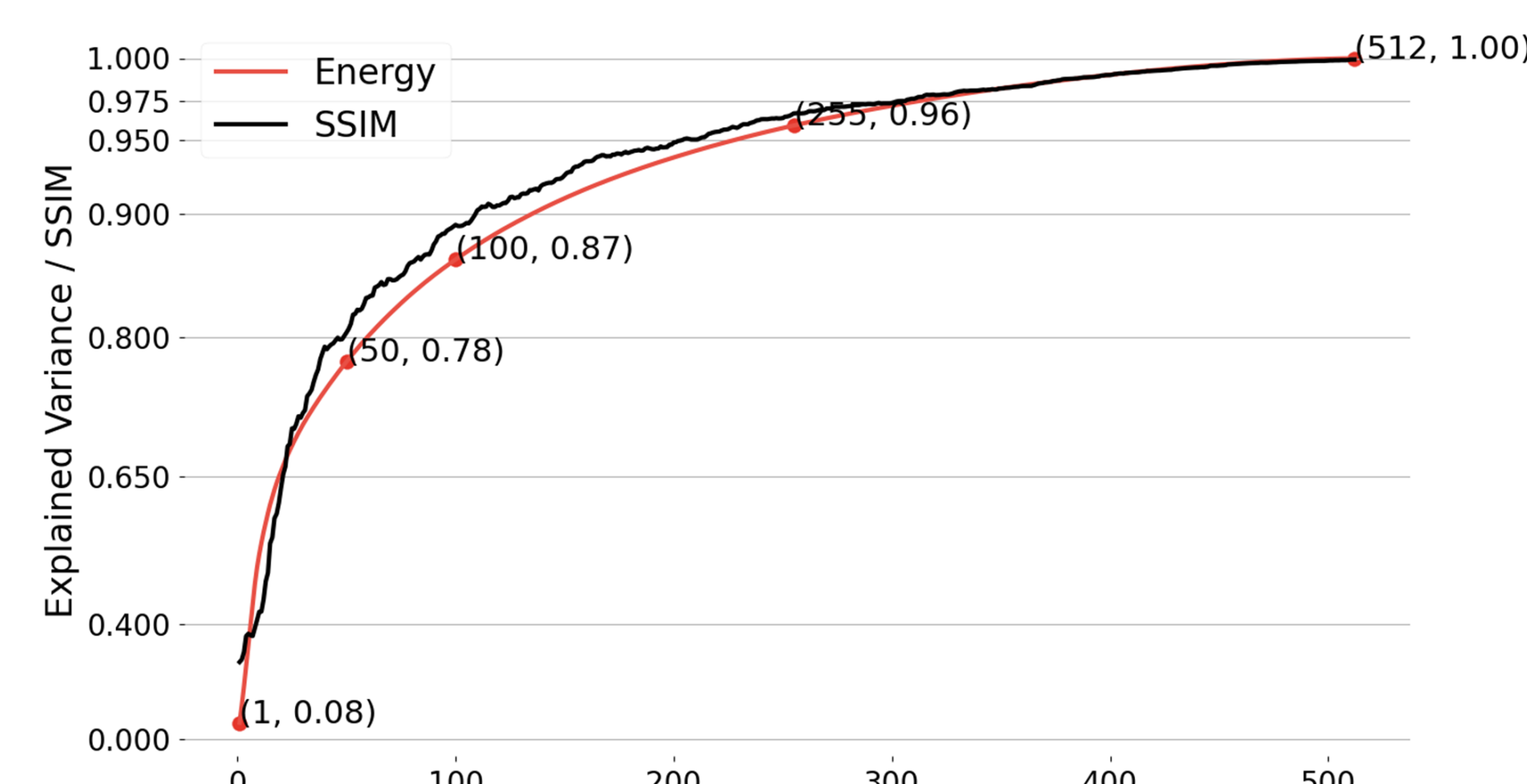
1. ABSTRACT

GANs tend to synthesize and manipulate global features, such as age, rather than focusing on local features such as facial wrinkles. Consequently, when a specific wrinkle is edited, all age-related features change as well. This paper proposes a new method that allows a GAN to **learn a specific disentangled edit**. The method involves **fine-tuning a pre-trained GAN using two discriminators**, each trained on a specific dataset representing distinct states of a single disentangled feature. Importantly, to avoid interfering with prior GAN knowledge, the **editing direction is defined in the meaningless dimension** of the GAN latent space.

2. CONTRIBUTIONS

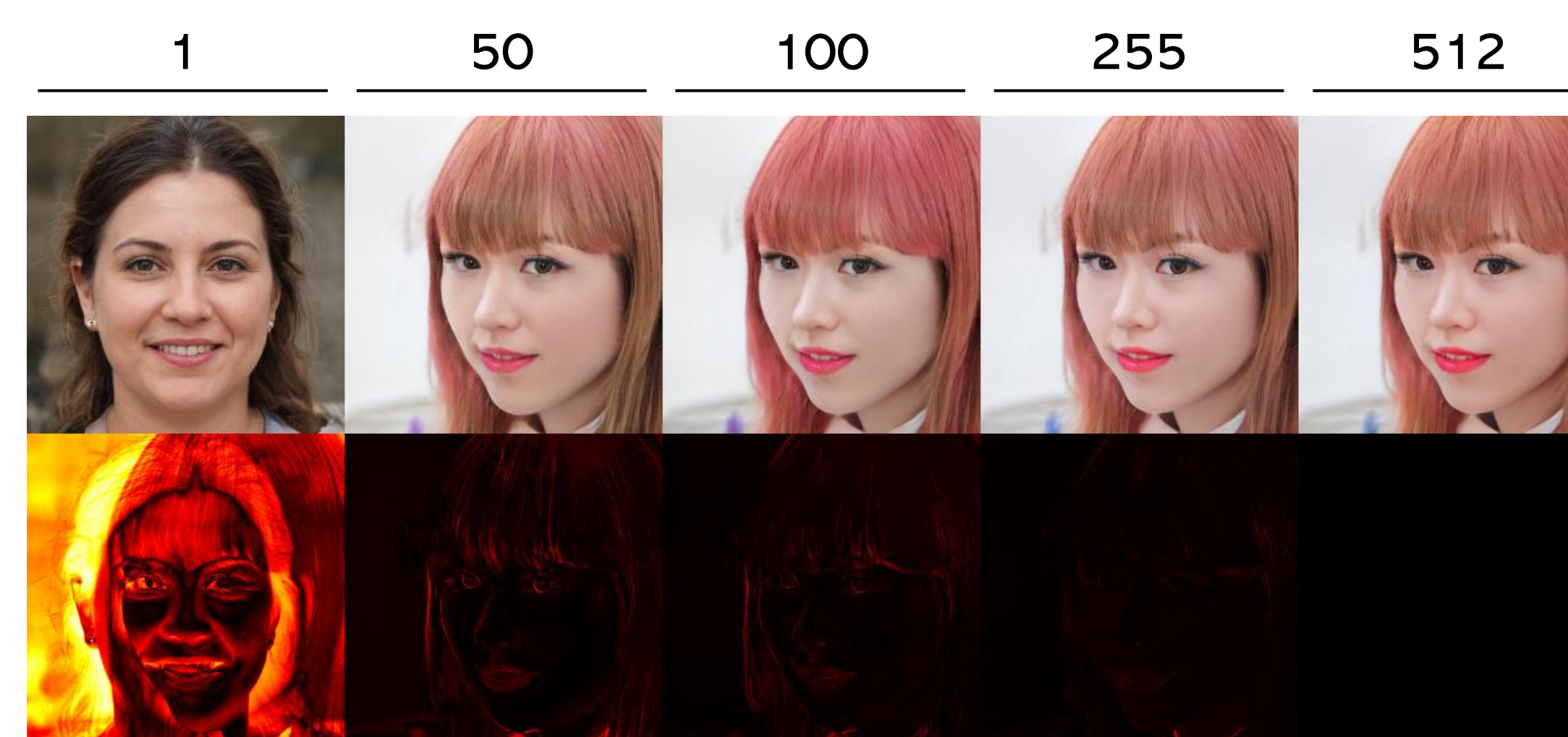
- We add a second discriminator to a pre-trained GAN to disentangle a given feature in the latent space. This is achieved by fine-tuning the GAN with two datasets, each representing a state of the feature to be edited.
- We identify the meaningless dimensions of the latent space, and create an editing direction in it to avoid interfering with the GAN's synthesis capacity
- We add fake images, created using traditional computer vision techniques, to the training dataset to disentangle highly correlated features.

3. METHOD

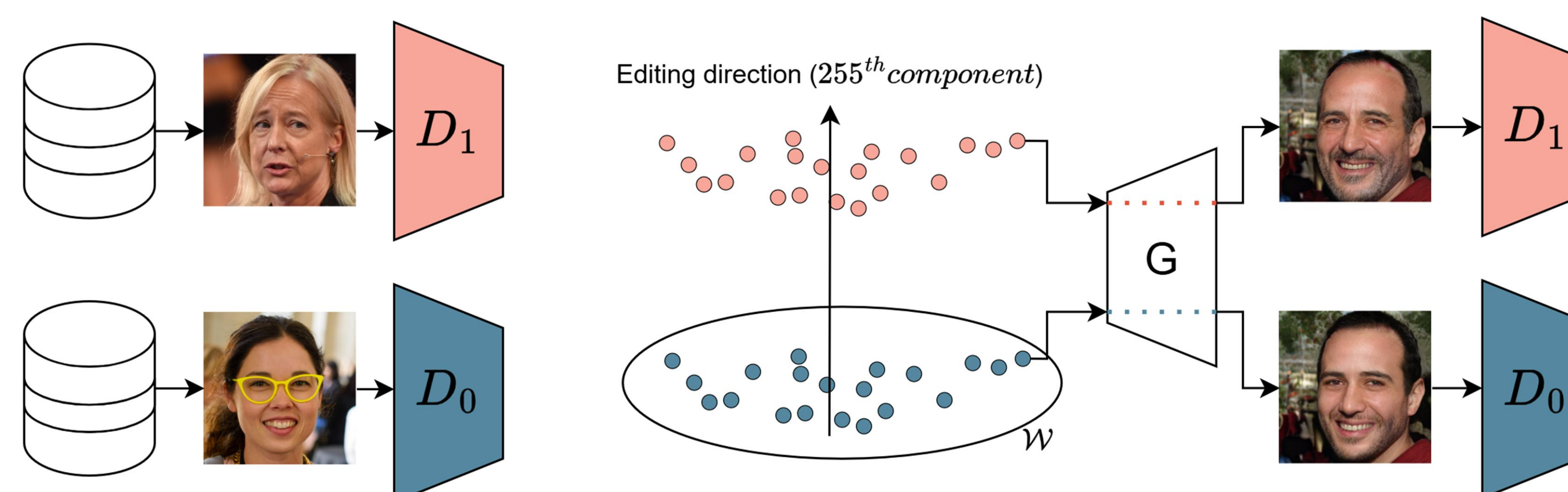


Only the **first principal components** (~ 100) of the latent space are sufficient to describe the **overall image appearance** [1]. This empiric observations are confirmed by the high correlation of energy with qualitative metrics such as MSE or SSIM.

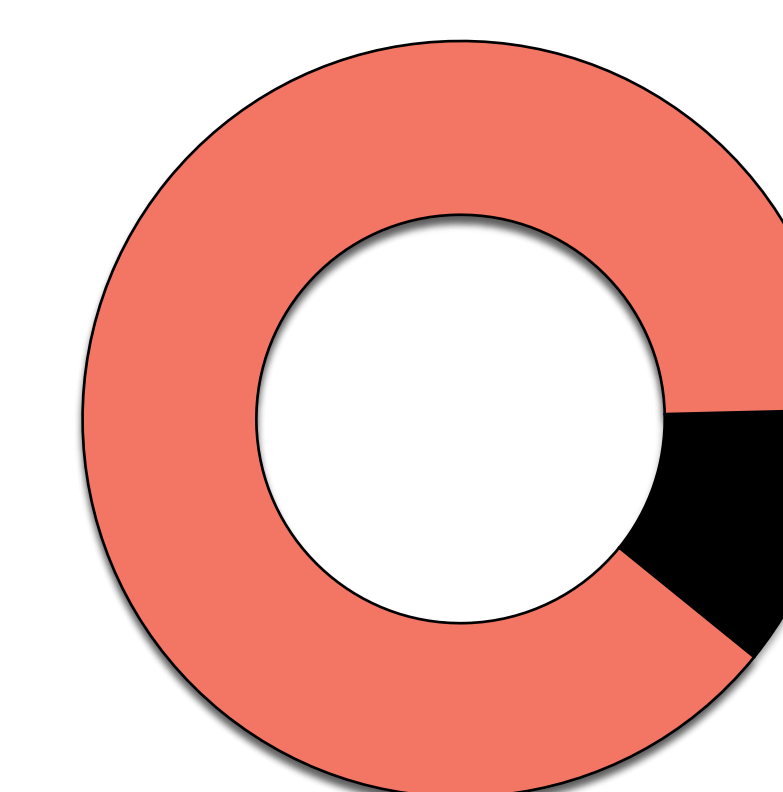
Based on these results, we propose using the **latest principal components** to define an editing direction in the latent space of a pre-trained GAN that **does not to interfere with its prior knowledge**.



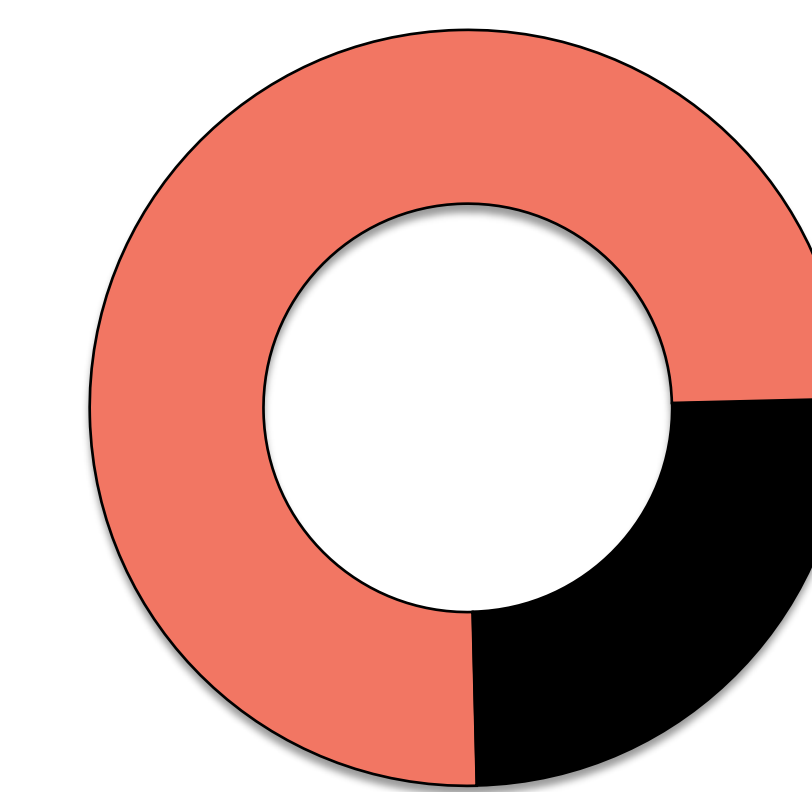
The generator is trained under the supervision of **two discriminators**. Each discriminator is **trained on a dataset corresponding to a particular state of the feature to be edited**. For example, for forehead wrinkle editing, the datasets are people without forehead wrinkles and people with forehead wrinkles. Each generated image is then fed into the appropriate discriminator based on its target features.



4. RESULTS



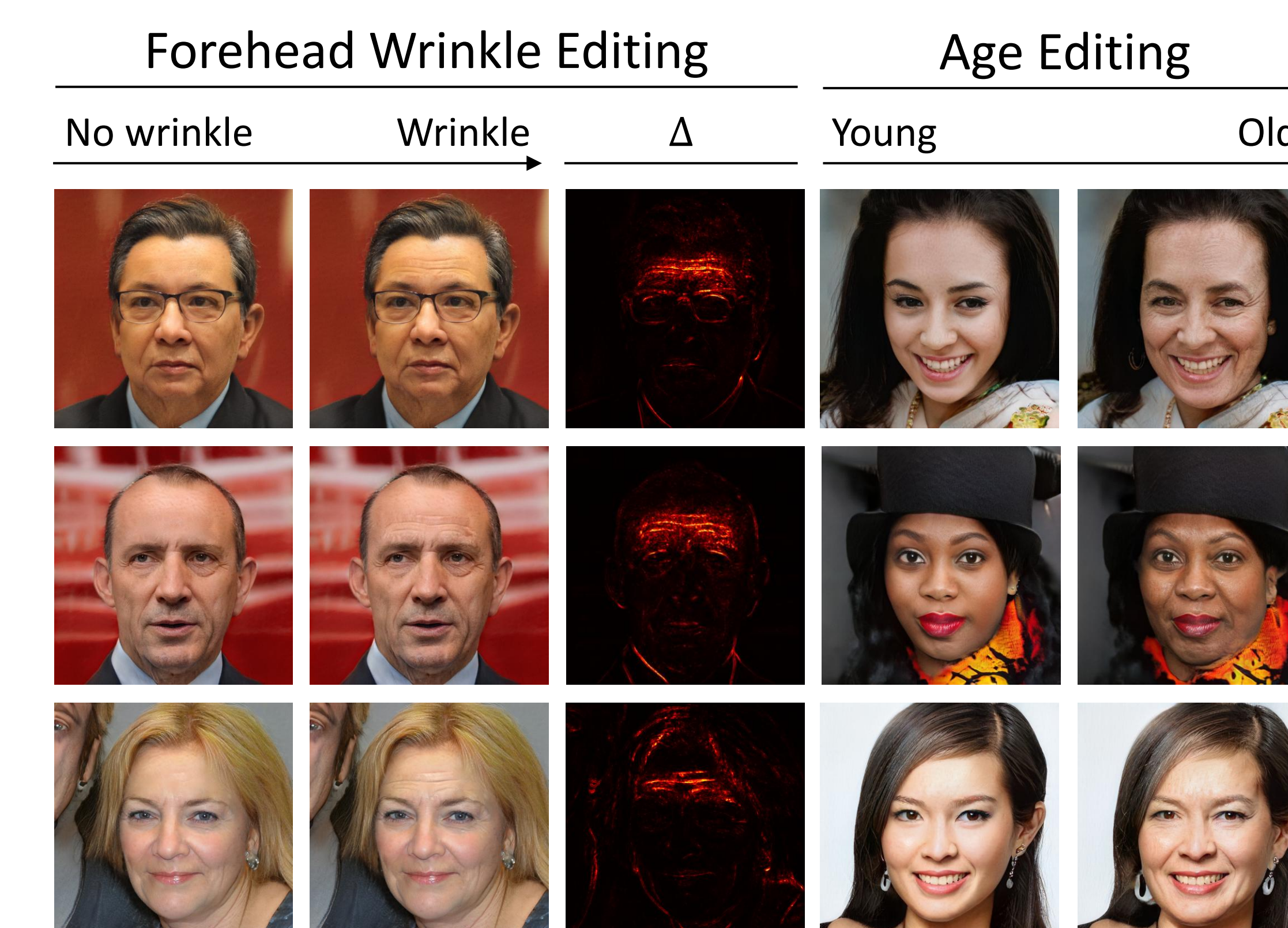
88% of accuracy for age editing



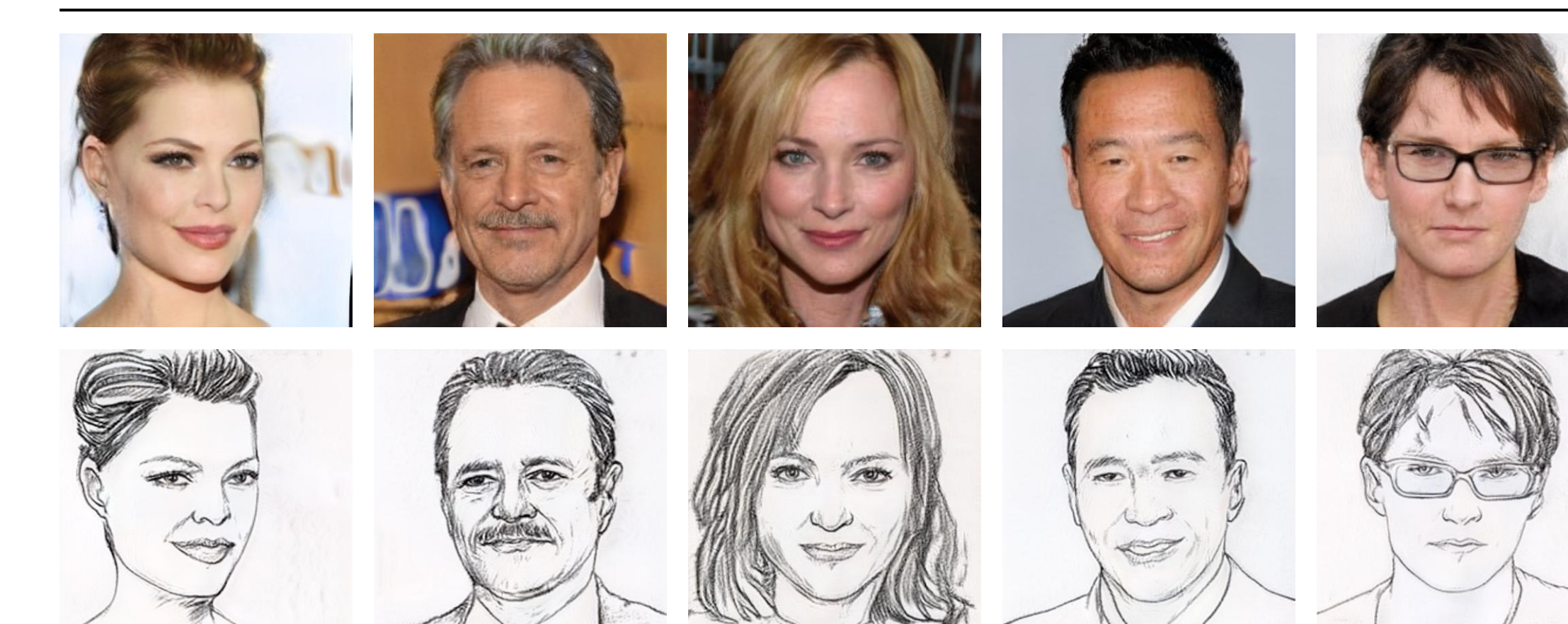
75% of accuracy for forehead wrinkle editing



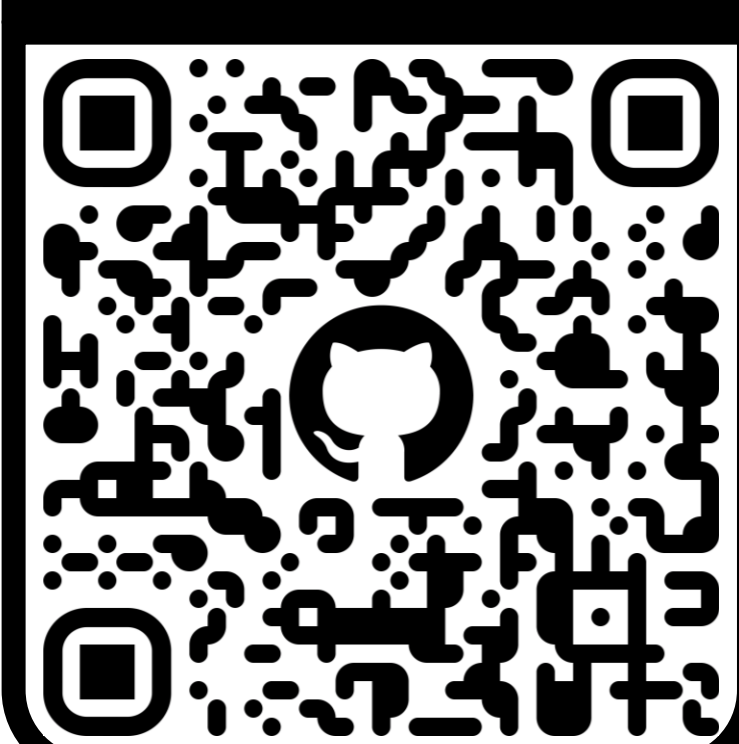
83% of people chose JanusGAN in a user study comparing it against a state-of-the-art method



Face to Sketch



Github Link



REFERENCES

- [1] E. Härkönen, A. Hertzmann, J. Lehtinen, and S. Paris. Ganspace: Discovering interpretable gan controls. In Proc. NeurIPS, 2020.
- [2] J. Choi and B. Han. Mcl-gan: Generative adversarial networks with multiple specialized discriminators. Advances in Neural InformationProcessing Systems, 35:29597–29609, 2022.